

Lagoa Spring 2024

EE 350

Test 3

Closed Book, Closed Notes, 3 Sheets $8\frac{1}{2}$ in $\times$ 11 in

Write Clearly and Fully Justify Your Answers

Question 1 (30%):

- a) (8%) The complex Fourier series of a function $f(t)$ of period $T_0 = 2\pi/\omega_0$ is of the form

$$f(t) = \sum_{n=-\infty}^{n=\infty} D_n e^{jn\omega_0 t}.$$

Consider a real periodic function $f(t)$ with period $T_0 = 2\pi$. The only nonzero coefficients of the complex Fourier series for $n \geq 0$ are $D_1 = 3j$, $D_3 = -10j$ and $D_5 = j$. Determine D_n for $n < 0$.

- b) (7%) Is the function even, odd or neither. Justify your answer.

If you have not done part a) use $D_1 = 3+j$, $D_{-1} = 3-j$, $D_2 = -10+2j$, $D_{-2} = -10-2j$, $D_3 = D_{-3} = 1$ and $D_n = 0$ for the remainder values of n .

- c) (7%) Determine the coefficients of the compact trigonometric series of $f(t)$; i.e., determine C_n and θ_n such that

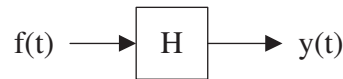
$$f(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n).$$

If you have not done part a) use $D_1 = 3+j$, $D_{-1} = 3-j$, $D_2 = -10+2j$, $D_{-2} = -10-2j$, $D_3 = D_{-3} = 1$ and $D_n = 0$ for the remainder values of n .

- d) (8%) Assume that a signal $f(t)$ of the form

$$f(t) = \sum_{n=-\infty}^{n=\infty} D_n e^{jn\omega_0 t}.$$

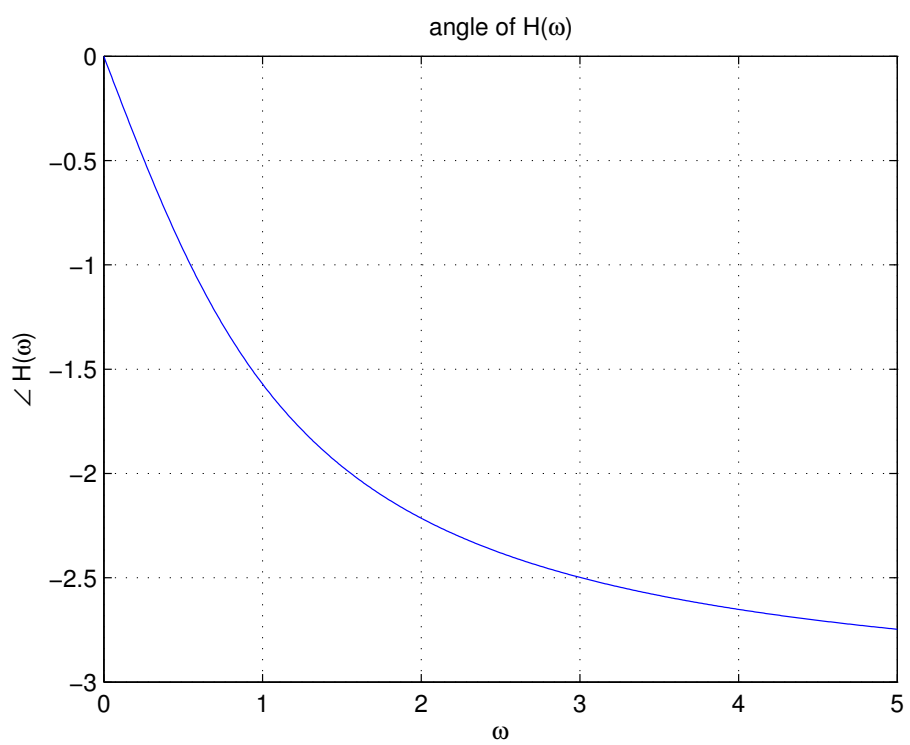
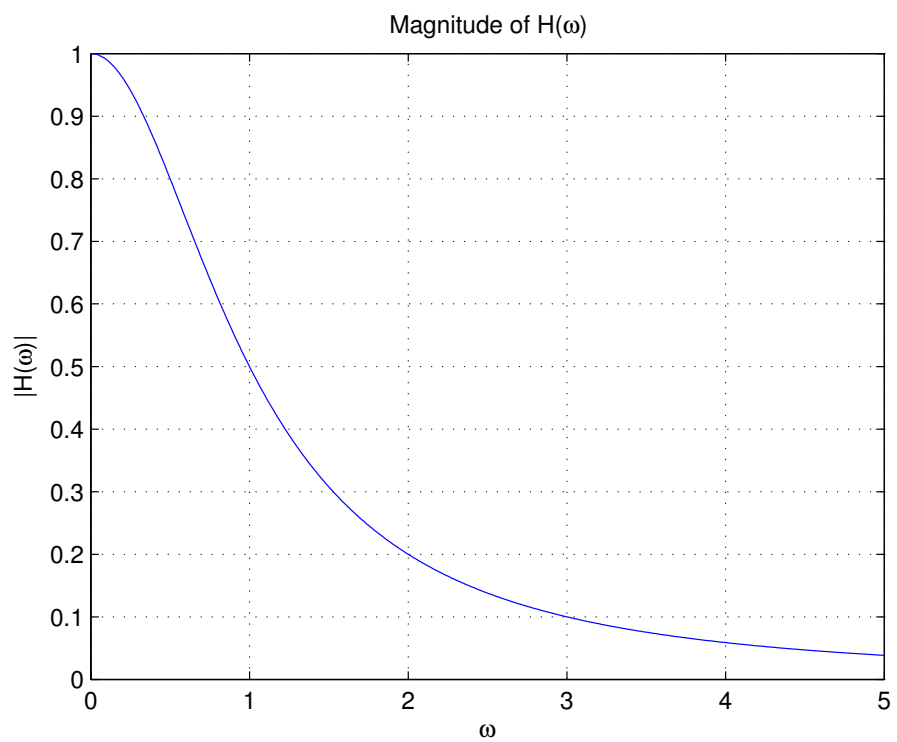
is filtered using a filter whose frequency response is given in the next page.



Let

- P_f be the power of the signal $f(t)$
- P_y be the power of the signal $y(t)$

Is $P_f \geq P_y$, $P_f \leq P_y$ or neither? Fully justify your answer.



Question 2 (35%):

Consider a system with an impulse response whose plot of its Fourier Transform is given in the next page.

- a) (9%) Assume that the impulse response of the system is of the form

$$h(t) = \alpha e^{-\beta t} u(t)$$

As mentioned above, the plot of the Fourier transform of $h(t)$ (denoted by $H(\omega)$) is given in the next page. Determine the values of α and β .

- b) (9%) Determine the response of the system to the input

$$f(t) = 2e^{-10t} u(t - 1).$$

If you have not done part a) assume that the Fourier transform of the impulse response of the system is

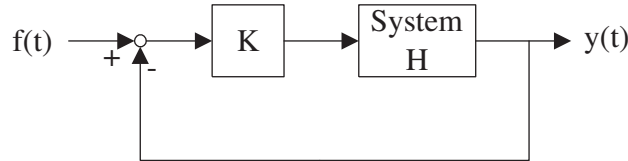
$$H(\omega) = \frac{1}{j\omega + 10}.$$

- c) (8%) Determine the differential equation that describes the system.

If you have not done part a) assume that the Fourier transform of the impulse response of the system is

$$H(\omega) = \frac{1}{j\omega + 10}.$$

- d) (9%) Consider the closed-loop system below



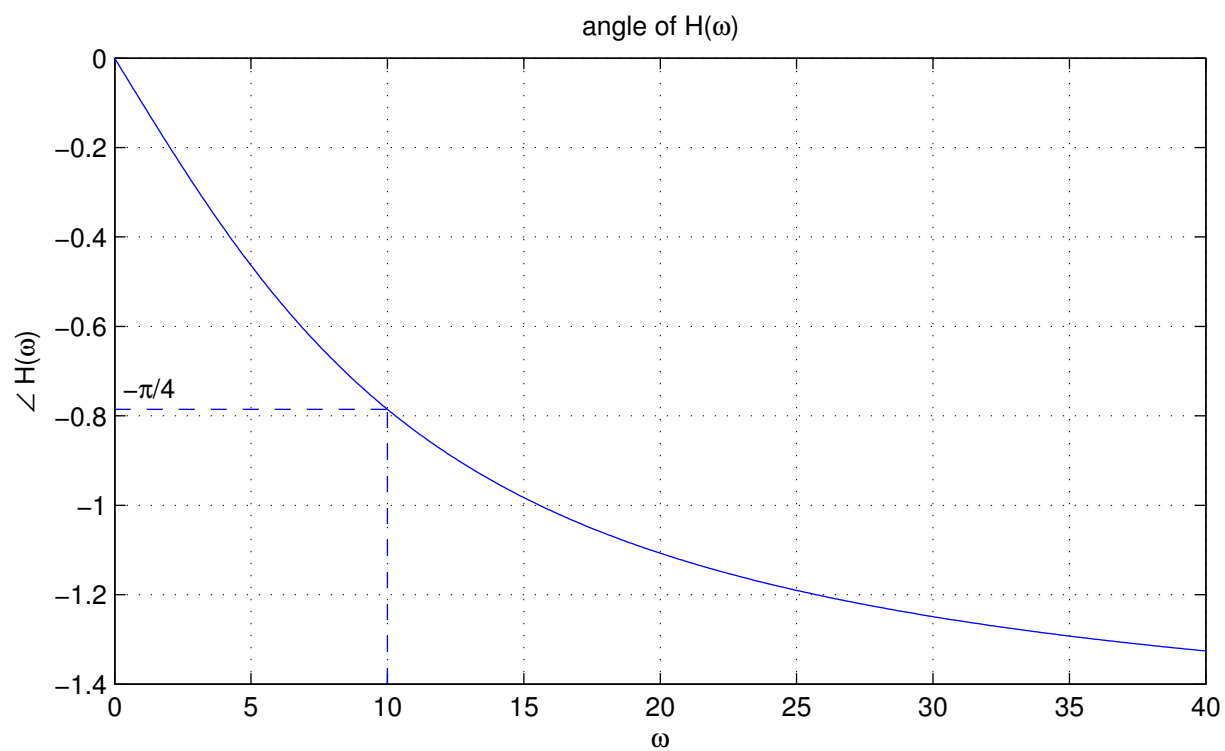
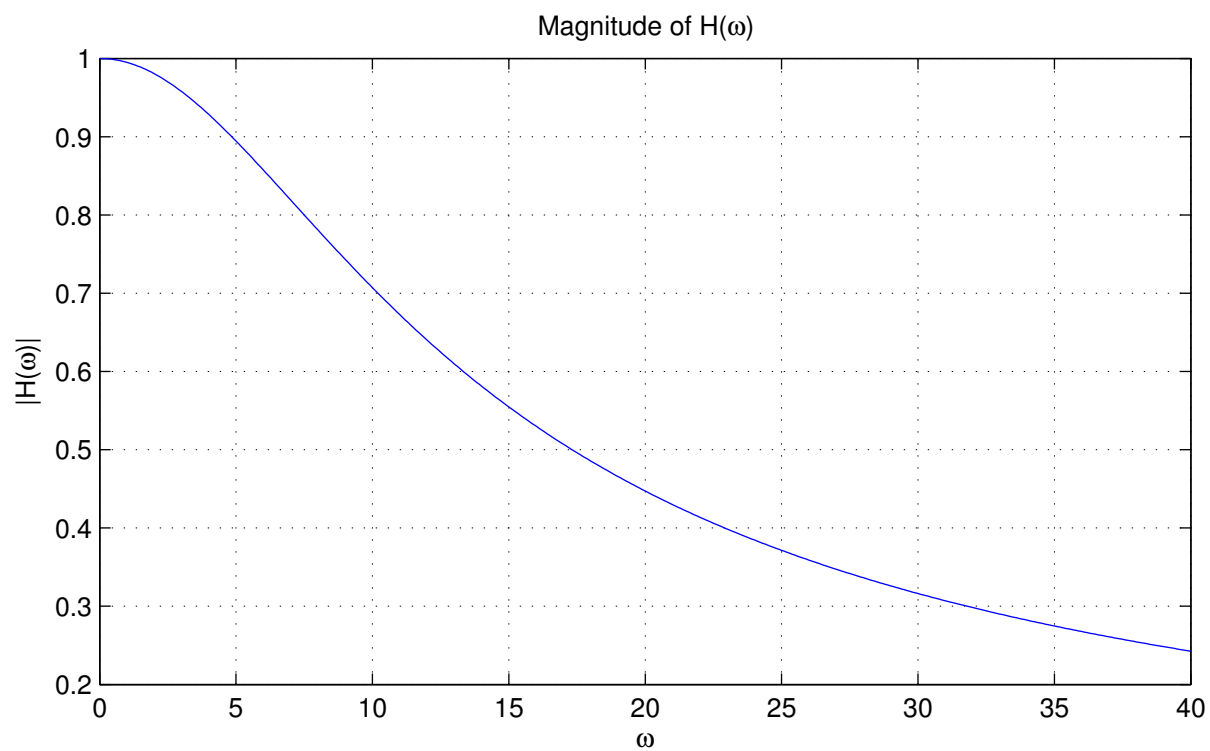
where System H is the one in part a) and K is a constant gain. When the input to the closed loop system is

$$f(t) = 2\sqrt{2} \cos(2t + \pi/2)$$

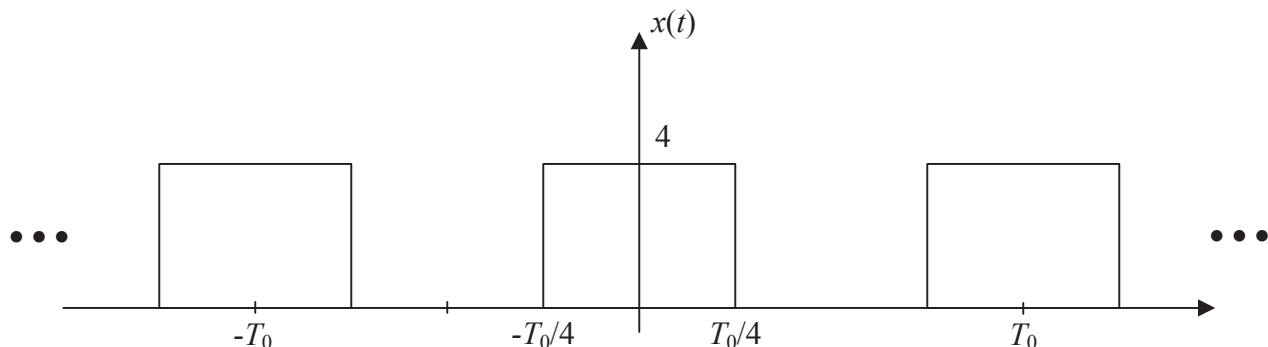
one obtains the following steady state output

$$y_{SS}(t) = \cos(2t + \pi/4).$$

What is the value of K ?



Question 3 (35%): Throughout this question $x(t)$ is the signal depicted below which is periodic with period T_0 .



- a) (9%) Find the complex Fourier series of the signal $x(t)$.
- b) (8%) Denoting by D_n the coefficients of the complex Fourier series computed above, recall that we have

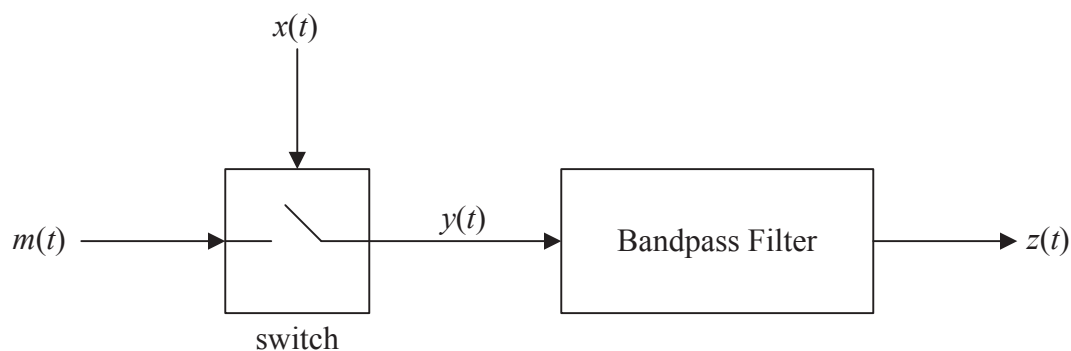
$$x(t) = \sum_{n=-\infty}^{+\infty} D_n e^{jn\omega_0 t} \quad \text{where} \quad \omega_0 = \frac{2\pi}{T_0}.$$

Find the Fourier transform of $x(t)$.

If you have not done part a) take

$$D_n = 3 \frac{\sin\left(\frac{n\pi}{3}\right)}{n\pi}$$

- c) (9%) Because analog multiplication is difficult to implement over a wide dynamical range, in AM modulation sometimes the switching scheme below is used to obtain the modulated signal.



where $x(t)$ is the signal above, $T_0 = \frac{2\pi}{\omega_c}$ and

$$y(t) = m(t)x(t).$$

Find the Fourier transform of $y(t)$ as a function of the Fourier transform of $m(t)$.

If you have not done part a) take

$$D_n = 3 \frac{\sin\left(\frac{n\pi}{3}\right)}{n\pi}$$

for $x(t)$.

d) (9%) Design the ideal bandpass filter above so that

$$z(t) = Km(t) \cos(\omega_c t)$$

where K is a constant. Assume that the bandwidth of the signal $m(t)$ is much smaller than ω_c .

If you have not done part a) take

$$D_n = 3 \frac{\sin\left(\frac{n\pi}{3}\right)}{n\pi}$$

for $x(t)$.